

LIMITS OF FUNCTIONS

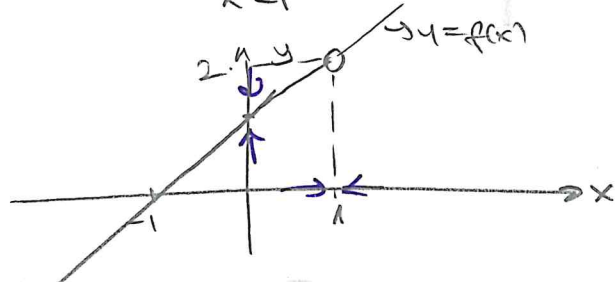
Ex Describe the behaviour of the functions near 1.

(a) $f(x) = \frac{x^2-1}{x-1}$

(b) $g(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1 \\ 1 & x = 1 \end{cases}$

(c) $h(x) = x+1$

(a) $f(x) = \frac{x^2-1}{x-1} = x+1$

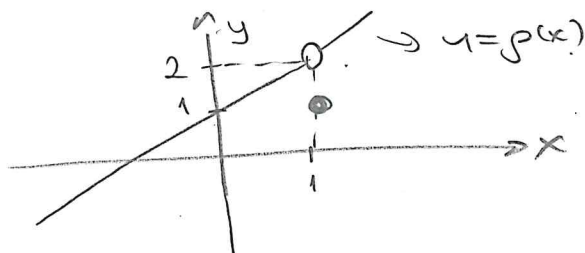


x	y
1.01	2.01
1.001	2.001
1.0001	2.0001
0.999	1.999
0.99	1.99
0.9	1.9

$\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$

NOTE Please note that the function f is not defined at $x=1$

(b) $g(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1 \\ 1 & x = 1 \end{cases}$

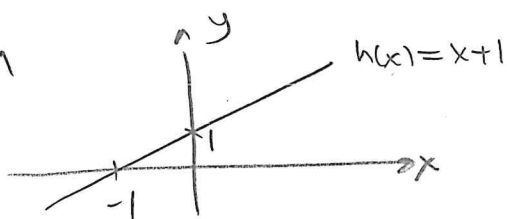


function is defined at $x=1$ $g(1)=1$

$\lim_{x \rightarrow 1} g(x) = 2$

$\lim_{x \rightarrow 1} g(x) = 2 \neq g(1) = 1$

(c) $h(x) = x+1$



$\lim_{x \rightarrow 1} x+1 = 2$

function is defined at $x=1$

$\lim_{x \rightarrow 1} h(x) = h(1)$

An informal definition of limit

If $f(x)$ is defined for all x near a , except possibly at a itself, and if we can ensure that $f(x)$ is as close as we want to L by taking x close enough to a , but not equal to a , but not equal to a , we say that the function f approaches the limit L as x approaches a we write

$$\lim_{x \rightarrow a} f(x) = L$$

Ex Evaluate the following limits

(a) $\lim_{x \rightarrow a} x = ?$ (b) $\lim_{x \rightarrow a} c = ?$ (c) $\lim_{x \rightarrow 2} g(x) = ?$ $g(x) = \begin{cases} x & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

(d) $\lim_{x \rightarrow -2} \frac{x^2 + x - 2}{x^2 + 5x + 6}$ (e) $\lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$ (f) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x^2 - 16}$

(a) $\lim_{x \rightarrow a} x = \boxed{a}$ (b) $\lim_{x \rightarrow a} c = \boxed{c}$ (c) $\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} x = \boxed{2}$
 $g(x) = x$ $g(a) = a$ $g(x) = c$ $g(a) = c$ $g(2) = 1$

(d) $\lim_{x \rightarrow -2} \frac{x^2 + x - 2}{x^2 + 5x + 6} = \lim_{x \rightarrow -2} \frac{(x+2)(x-1)}{(x+2)(x+3)} = \boxed{-\frac{3}{5}}$

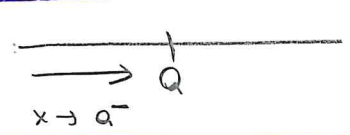
(e) $\lim_{x \rightarrow a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a} = \lim_{x \rightarrow a} \frac{\frac{a-x}{ax} \cdot \frac{1}{x-a}}{x-a} = \boxed{-\frac{1}{a^2}}$

(f) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x^2 - 16} = \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{(x-4)(x+4)} = \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{(\sqrt{x} - 2)(\sqrt{x} + 2)(x+4)} = \boxed{\frac{1}{32}}$

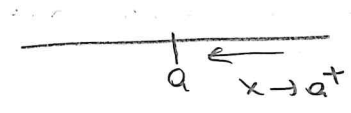
ONE SIDED LIMITS

Informal definition of left and right limits

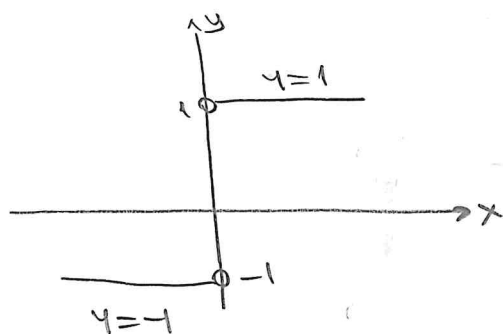
→ If $f(x)$ is defined on some interval (b, a) extending to the left of $x=a$, and if we can ensure that $f(x)$ is as close as we want to L by taking x to the left of a and close enough to a , then we say $f(x)$ has left limit L at $x=a$ and write

$$\lim_{x \rightarrow a^-} f(x) = L$$


→ If $f(x)$ is defined on some interval (a, b) extending to the right of $x=a$, and if we can ensure that $f(x)$ is as close as we want to L by taking x to the right of a and close enough to a , then we say $f(x)$ has right limit L at $x=a$ and write

$$\lim_{x \rightarrow a^+} f(x) = L$$


Ex



$$y = \text{sign}(x) = \frac{x}{|x|} \text{ function.}$$

$$\lim_{x \rightarrow 0^-} \text{sign}(x) = -1 \quad \text{limit doesn't exist.}$$

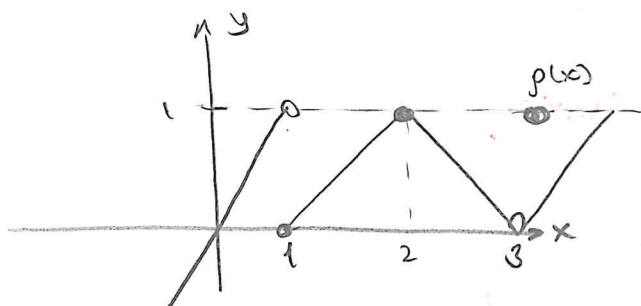
$$\lim_{x \rightarrow 0^+} \text{sign}(x) = 1$$

Thm Relationship between one-sided and two-sided limits

A function $f(x)$ has limit L at $x=a$ if and only if it has both left and right limits there and these one-sided limits are both equal to L :

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

Ex 7



$$\lim_{x \rightarrow 1} g(x) = ?$$

$$\lim_{x \rightarrow 2} g(x) = ?$$

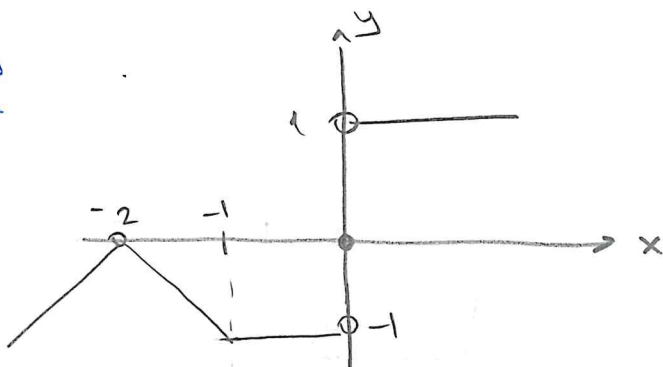
$$\lim_{x \rightarrow 3} g(x) = ?$$

$$\lim_{x \rightarrow 1^+} g(x) = 0 \quad \lim_{x \rightarrow 1^-} g(x) = 1 \quad 0 \neq 1 \quad \text{limit does not exist}$$

$$\lim_{x \rightarrow 2^+} g(x) = 1 \quad \lim_{x \rightarrow 2^-} g(x) = 1 \quad \Rightarrow \lim_{x \rightarrow 2} g(x) = 1$$

$$\lim_{x \rightarrow 3^+} g(x) = 0 \quad \lim_{x \rightarrow 3^-} g(x) = 0 \quad \Rightarrow \lim_{x \rightarrow 3} g(x) = 0$$

Ex 7



$$\lim_{x \rightarrow -2} f(x) = ?$$

$$\lim_{x \rightarrow -1} f(x) = ?$$

$$\lim_{x \rightarrow 0} f(x) = ?$$

$$\left. \begin{array}{l} \lim_{x \rightarrow 0^+} f(x) = 1 \\ \lim_{x \rightarrow 0^-} f(x) = -1 \end{array} \right\} 1 \neq -1 \quad \lim_{x \rightarrow 0} f(x)$$

$$\left. \begin{array}{l} \lim_{x \rightarrow -1^-} f(x) = -1 \\ \lim_{x \rightarrow -1^+} f(x) = -1 \end{array} \right\} \lim_{x \rightarrow -1} f(x) = -1$$

$$\lim_{x \rightarrow -2^+} f(x) = 0 \quad \left\{ \lim_{x \rightarrow -2} f(x) = 0 \right.$$

$$\lim_{x \rightarrow -2^-} f(x) = 0$$

Ex 7 $f(x) = \frac{x-2}{x^2+x-6}$

$$\lim_{x \rightarrow 2^+} f(x) = ?$$

$$\lim_{x \rightarrow 2^-} f(x) = ?$$

$$\lim_{x \rightarrow 2} f(x) = ?$$

$$x > 2 \quad \lim_{x \rightarrow 2^+} \frac{x-2}{x^2+x-6} = \lim_{x \rightarrow 2^+} \frac{x-2}{(x-2)(x+3)} = \frac{1}{5}$$

$$\left\{ \frac{1}{5} \neq -\frac{1}{5} \right.$$

$$\lim_{x \rightarrow 2} f(x)$$

$$x < 2 \quad \lim_{x \rightarrow 2^-} \frac{x-2}{x^2+x-6} = \lim_{x \rightarrow 2^-} \frac{2-x}{(x-2)(x+3)} = -\frac{1}{5}$$

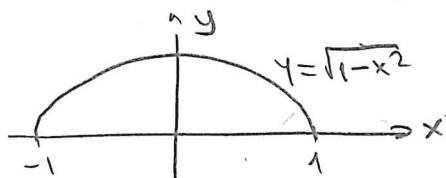
does not exist.

Ex 7

$$g(x) = \sqrt{1-x^2}$$

$$\lim_{x \rightarrow -1} g(x) = ?$$

$$\lim_{x \rightarrow 1} g(x) = ?$$



$$\bullet \lim_{x \rightarrow -1^+} g(x) = 0$$

no left limit at $x = -1$
no limit at $x = -1$

$$\bullet \lim_{x \rightarrow 1^-} g(x) = 0$$

no right limit at $x = 1$
no limit at $x = 1$

RULES FOR CALCULATING LIMITS

If $\lim_{x \rightarrow a} f(x) = L$, $\lim_{x \rightarrow a} g(x) = M$ and $L \in \mathbb{R}$, then

- ① $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$
- ② $\lim_{x \rightarrow a} (f(x) - g(x)) = L - M$
- ③ $\lim_{x \rightarrow a} f(x)g(x) = L \cdot M$
- ④ $\lim_{x \rightarrow a} k f(x) = kL$
- ⑤ $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M} \quad M \neq 0$

If n is an integer and n is a positive integer, then

$$\textcircled{6} \lim_{x \rightarrow a} [f(x)]^{n/n} = L^{n/n} \quad \text{provided } \begin{array}{l} L > 0 \text{ if } n \text{ is even} \\ L \neq 0 \text{ if } n < 0 \end{array}$$

If $f(x) \leq g(x)$ on an interval containing a in its interior, then

$$\textcircled{7} L \leq M \quad (\text{order preserved})$$

Thm: Limits of Polynomials and Rational Functions

$$\textcircled{1} \text{ If } P(x) \text{ is a polynomial and } a \text{ is any real number, then}$$

$$\lim_{x \rightarrow a} P(x) = P(a)$$

$$\textcircled{2} \text{ If } P(x) \text{ and } Q(x) \text{ are polynomials and } Q(a) \neq 0, \text{ then}$$

$$\lim_{x \rightarrow a} \frac{P(x)}{Q(x)} = \frac{P(a)}{Q(a)}$$

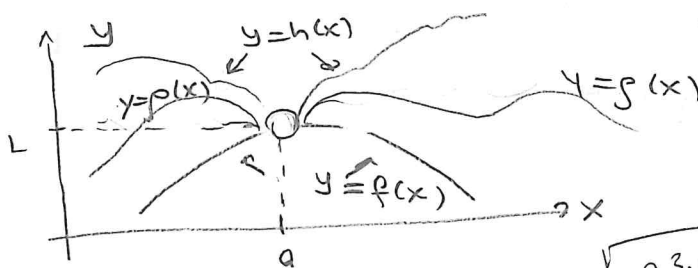
Thm The squeeze (or sandwich) theorem

Suppose that $f(x) \leq g(x) \leq h(x)$ holds for all x in some open interval containing a , except possibly at $x=a$ itself.

Suppose also that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

Then, $\lim_{x \rightarrow a} g(x) = L$ also. Similar statements hold for left and right limits.



$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Examples (1) $\lim_{y \rightarrow 2} \frac{y+2}{y^2+5y+6} = \lim_{y \rightarrow 2} \frac{y+2}{(y+2)(y+3)} = \frac{1}{5}$

(2) $\lim_{x \rightarrow -5} \frac{x^2+3x-10}{x+5} = \lim_{x \rightarrow -5} \frac{(x+5)(x-2)}{(x+5)} = -7$

(3) $\lim_{u \rightarrow 1} \frac{u^4-1}{u^3-1} = \lim_{u \rightarrow 1} \frac{(u^2-1)(u^2+1)}{(u-1)(u^2+u+1)} = \lim_{u \rightarrow 1} \frac{(u-1)(u+1)(u^2+1)}{(u-1)(u^2+u+1)} = \frac{4}{3}$

(4) $\lim_{x \rightarrow 9} \frac{\sqrt{x}-3}{x-9} = \lim_{x \rightarrow 9} \frac{\sqrt{x}-3}{(\sqrt{x}-3)(\sqrt{x}+3)} = \frac{1}{6}$

(5) $\lim_{x \rightarrow -1} \frac{\sqrt{x^2+8}-3}{x+1} = \lim_{x \rightarrow -1} \frac{(\sqrt{x^2+8}-3)(\sqrt{x^2+8}+3)}{(x+1)(\sqrt{x^2+8}+3)}$

$$= \lim_{x \rightarrow -1} \frac{(x^2+8-9)}{(x+1)(\sqrt{x^2+8}+3)} = \lim_{x \rightarrow -1} \frac{(x-1)}{(x+1)(\sqrt{x^2+8}+3)} = -\frac{2}{6}$$

(6)

6) $\lim_{x \rightarrow 0} \frac{\frac{1}{x} - 1}{x - a} = \lim_{x \rightarrow a} \frac{\frac{a-x}{a \cdot x}}{x-a} = \frac{-1}{a^2}$

(7) $\lim_{x \rightarrow 0} \frac{|x|}{x} = ?$ $\lim_{x \rightarrow 0^+} \frac{x}{x} = 1$ $\lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$ $1 \neq -1$ LIMIT DOES NOT EXIST.

8) $\lim_{h \rightarrow 0} \frac{\sqrt{1+h} - 1}{h} = \lim_{h \rightarrow 0} \frac{(\sqrt{1+h} - 1)(\sqrt{1+h} + 1)}{h} = \lim_{h \rightarrow 0} \frac{1+h-1}{h} = 1$

$$\begin{aligned} \textcircled{9} \quad \lim_{t \rightarrow 0} \frac{1}{t\sqrt{1+t}} - \frac{1}{t} &= \lim_{t \rightarrow 0} \frac{1 - \sqrt{1+t}}{t\sqrt{1+t}} = \lim_{t \rightarrow 0} \frac{1 - (1+t)}{t(\sqrt{1+t})(1 + \sqrt{1+t})} \\ &= \lim_{t \rightarrow 0} \frac{-t}{t(\sqrt{1+t})(1 + \sqrt{1+t})} = \frac{-1}{2} \end{aligned}$$

(10) Given that $3 - x^2 \leq u(x) \leq 3 + x^2$ for all $x \neq 0$ find $\lim_{x \rightarrow 0} u(x) = ?$

$$\lim_{x \rightarrow 0} 3 - x^2 = 3 \quad \lim_{x \rightarrow 0} 3 + x^2 = 3 \quad \text{By Squeeze Theorem}$$

$$\lim_{x \rightarrow 7} u(x) = 3$$

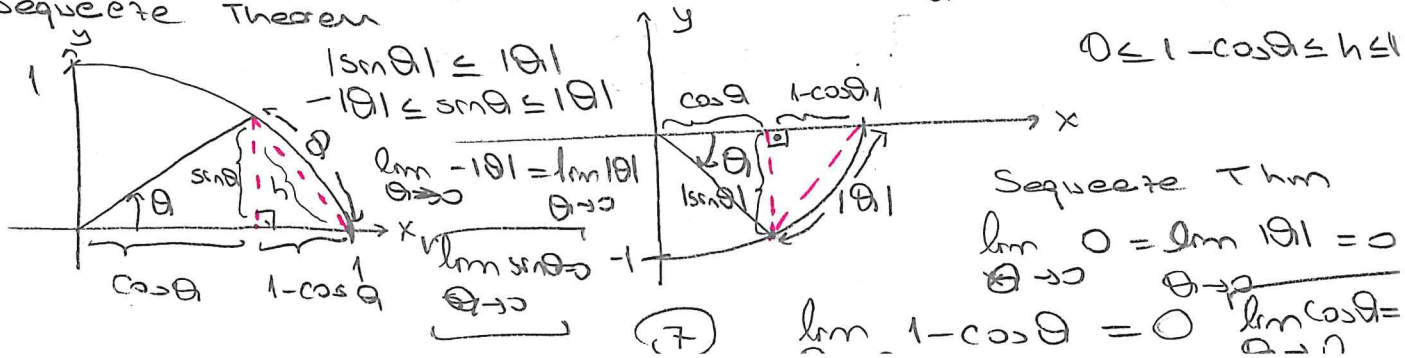
11 Show that $\lim_{x \rightarrow a} |f(x)| = 0$, then $\lim_{x \rightarrow a} f(x) = 0$.

Since $-|f(x)| \leq f(x) \leq |f(x)|$ $-|f(x)|$ and $|f(x)|$ both have limit 0 as x approaches to 0. Therefore, by Squeeze Theorem

$$\lim_{x \rightarrow \infty} f(x) = 0$$

(12) Show that $\lim_{\theta \rightarrow 0} \cos \theta = 1$ and $\lim_{\theta \rightarrow 0} \sin \theta = 0$. By using

Sequente Theorem

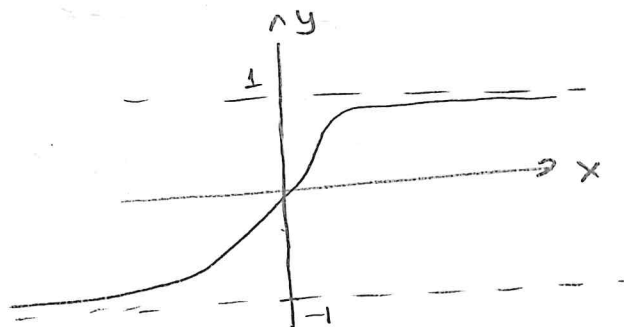


Limits at Infinity and Infinite Limits

Ex Consider $f(x) = \frac{x}{\sqrt{x^2+1}}$

Graph of $f(x) = \frac{x}{\sqrt{x^2+1}}$

x	f(x)
-1000	-0.9999995
-100	-0.9999500
-10	-0.9950372
-1	-0.7071068
0	0
1	0.7071068
10	0.9950372
100	0.9999500
1000	0.9999995



$\lim_{x \rightarrow \infty} f(x) = 1$ "f(x) approaches 1 as x approaches infinity"

$\lim_{x \rightarrow -\infty} f(x) = -1$ "f(x) approaches -1 as x approaches negative infinity"

Definition

Limits at infinity and negative infinity (informal definition)

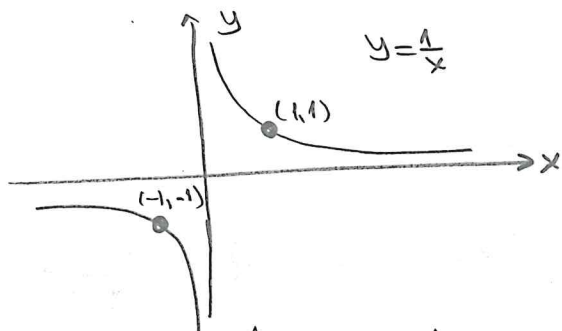
If the function f is defined on an interval (a, ∞) and if we ensure that $f(x)$ is as close as we want to the number L by taking x large enough, then we say that $f(x)$ approaches the limit L as x approaches infinity, and we write

$$\lim_{x \rightarrow \infty} f(x) = L$$

If f is defined on an interval $(-\infty, b)$ and if we can ensure that $f(x)$ is as close as we want to the number M by taking x negative and large enough in absolute value, then we say that $f(x)$ approaches the limit M as x approaches negative infinity, and we write

$$\lim_{x \rightarrow -\infty} f(x) = M$$

Ex ①



$$\lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0$$

② Evaluate $\lim_{x \rightarrow \infty} f(x)$, $\lim_{x \rightarrow -\infty} f(x)$ for $f(x) = \frac{x}{\sqrt{x^2+1}}$

$$f(x) = \frac{x}{\sqrt{x^2(1+\frac{1}{x^2})}} = \frac{x}{\sqrt{x^2} \sqrt{1+\frac{1}{x^2}}} = \frac{x}{|x| \sqrt{1+\frac{1}{x^2}}}$$

$\sqrt{x^2} = |x|$
 $\text{sgn } x = \frac{x}{|x|} = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$

$$f(x) = \frac{\text{sgn } x}{\sqrt{1+\frac{1}{x^2}}}$$

$$\lim_{x \rightarrow \infty} \sqrt{1+\frac{1}{x^2}} = 1$$

$$\lim_{x \rightarrow \infty} \frac{\text{sgn } x}{\sqrt{1+\frac{1}{x^2}}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{\text{sgn } x}{\sqrt{1+\frac{1}{x^2}}} = -1$$

Limits at Infinity for Rational Functions

Ex (Numerator and denominator of the same degree)

$$\lim_{x \rightarrow \pm\infty} \frac{2x^2 - x + 3}{3x^2 + 5} = \lim_{x \rightarrow \pm\infty} \frac{2 - \frac{1}{x} + \frac{3}{x^2}}{3 + \frac{5}{x^2}} = \frac{2}{3}$$

Ex (Degree of numerator less than degree of denominator)

$$\lim_{x \rightarrow \pm\infty} \frac{5x+2}{2x^3-1} = \lim_{x \rightarrow \pm\infty} \frac{\frac{5}{x^2} + \frac{2}{x^3}}{2 - \frac{1}{x^3}} = 0$$

Def Let $P_m(x) = a_m x^m + \dots + a_0$ and $Q_n(x) = b_n x^n + \dots + b_0$ be polynomials of degree m and n , respectively, so that $a_m \neq 0$ and $b_n \neq 0$. Then,

$$\lim_{x \rightarrow \pm\infty} \frac{P_m(x)}{Q_n(x)}$$

- (a) equals zero if $m < n$.
- (b) equals $\frac{a_m}{b_n}$ if $m = n$.
- (c) doesn't exist if $m > n$.

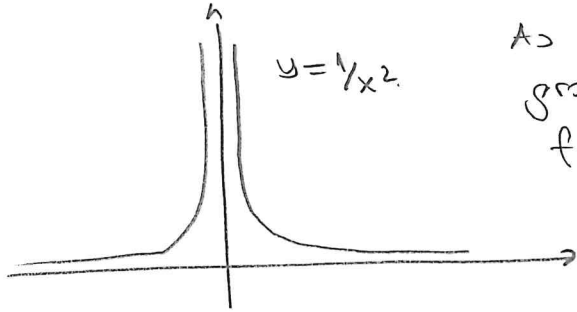
(a)

Ex 1 $\lim_{x \rightarrow \infty} (\sqrt{x^2+x} - x) = ?$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+x} - x}{(\sqrt{x^2+x} + x)} = \lim_{x \rightarrow \infty} \frac{x^2+x - x^2}{\sqrt{x^2+x} + x} = \lim_{x \rightarrow \infty} \frac{x}{x(\sqrt{1+\frac{1}{x}} + 1)} = \frac{1}{2}$$

Ex 2 (Two sided infinite limit)

Describe the behaviour of the function $f(x) = \frac{1}{x^2}$



As x approaches 0, $f(x)$ is positive and grows larger and larger. Therefore limit of $f(x)$ as x approaches 0 does not exist

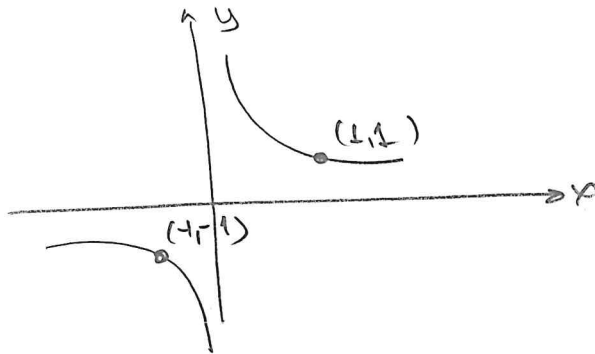
$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

Writing it we ARE NOT saying that $\lim_{x \rightarrow 0} \frac{1}{x^2}$ exists. Rather, we are saying that limit does not exist $\frac{1}{x^2}$ becomes arbitrary

large near $x=0$

Ex 3 (One-sided infinite limit) Describe the behaviour of the

function $f(x) = \frac{1}{x}$ near $x=0$



$$\lim_{x \rightarrow 0^+} f(x) = \infty$$

$$\lim_{x \rightarrow 0^-} f(x) = -\infty$$

Ex 4 (Polynomial Behaviour at infinity)

a $\lim_{x \rightarrow \infty} (3x^3 - x^2 + 2) = \infty$
 $= \lim_{x \rightarrow \infty} 3x^3 \left(1 - \frac{1}{3x} + \frac{2}{x^3}\right) = \infty$

c $\lim_{x \rightarrow \infty} (x^4 - 5x^3 - x) = \infty$

b $\lim_{x \rightarrow -\infty} (3x^3 - x^2 + 2) = -\infty$

d $\lim_{x \rightarrow -\infty} (x^4 - 5x^3 - x) = \infty$

Ex 5 (Rational functions with numerator of higher degree)

$$\lim_{x \rightarrow \infty} \frac{x^3+1}{x^2+1} = \lim_{x \rightarrow \infty} \frac{\cancel{x^3} (x + \cancel{1/x^2})}{\cancel{x^2} (1 + \cancel{1/x^2})} = \infty$$

Ex 6 (a) $\lim_{x \rightarrow 2} \frac{(x-2)^2}{x^2-4} = \lim_{x \rightarrow 2} \frac{\cancel{(x-2)} (x-2)}{\cancel{(x-2)} (x+2)} = \boxed{0}$

(b) $\lim_{x \rightarrow 2} \frac{x-2}{x^2-4} = \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}}{\cancel{(x-2)} (x+2)} = \boxed{1/4}$

(c) $\lim_{x \rightarrow 2^+} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{x-3}{(x-2)(x+2)} = -\infty$

$\frac{1}{2} \leftarrow 2^+$

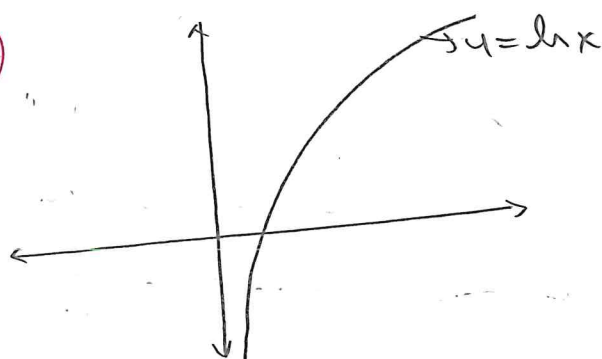
$\lim_{x \rightarrow 2^-} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^-} \frac{x-3}{(x-2)(x+2)} = \infty$

$\frac{1}{2^-}$

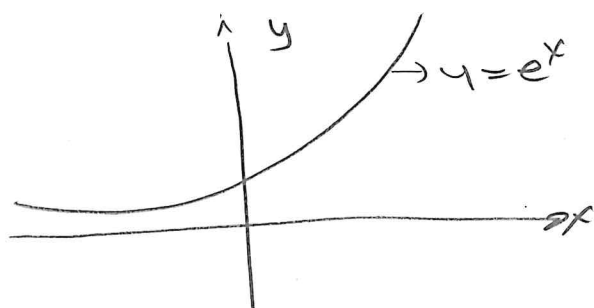
Therefore, $\lim_{x \rightarrow 2} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2} \frac{x-3}{(x-2)(x+2)}$ does not exist.

(d) $\lim_{x \rightarrow 2} \frac{2-x}{(x-2)^3} = \lim_{x \rightarrow 2} -\frac{(x-2)}{(x-2)^3} = \lim_{x \rightarrow 2} -\frac{1}{(x-2)^2} = -\infty$

(e) ***



$\lim_{x \rightarrow 0^+} f(x) = -\infty$



$\lim_{x \rightarrow \infty} e^x = \infty$

$\lim_{x \rightarrow -\infty} e^x = 0$

CONTINUITY

→ Suppose the domain of a function $f(x)$ is either an interval or a union of separate intervals. A point P in the domain of $f(x)$ is called an **interior point** if it belongs to some open interval contained in the domain. If it is not an interior point then it is called an **end point**.

Ex The function $f(x) = \sqrt{1-x^2}$ has domain $[-1, 1]$. The interior points forms up the open interval $(-1, 1)$. The endpoints are -1 and 1 .

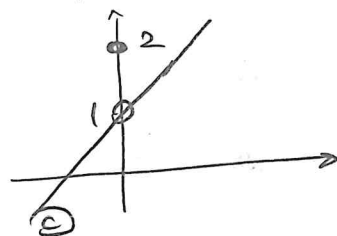
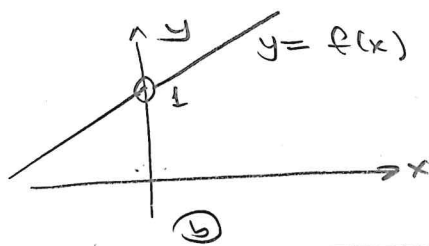
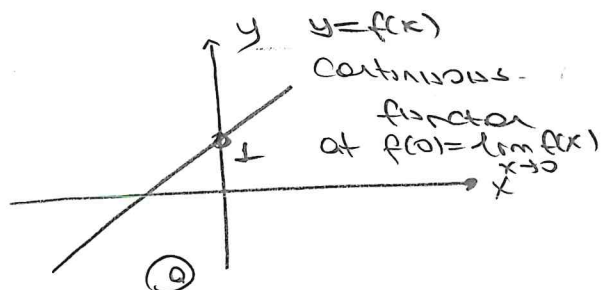
Ex The domain of the function $g(x) = \frac{1}{x}$ is the union of the open intervals $(-\infty, 0) \cup (0, \infty)$ and consists entirely of interior points.

Continuity at an interior point

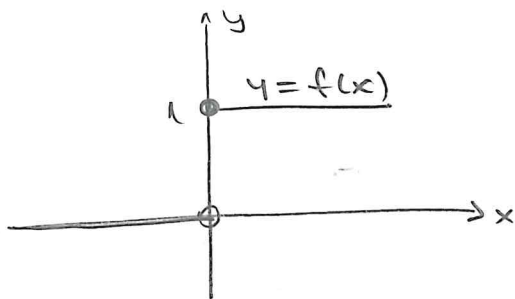
→ We say that a function f is continuous at an interior point c of its domain if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

If either $\lim_{x \rightarrow c} f(x)$ fails to exist or it exists but is not equal to $f(c)$, then we will say that f is discontinuous at c .

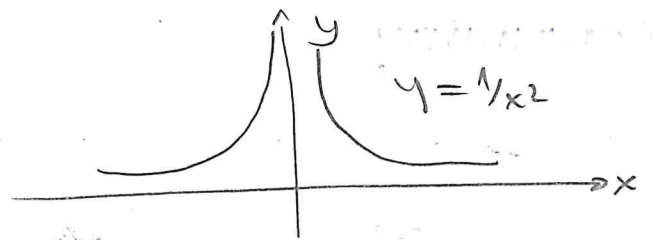


- Discontinuous at $x=0$
- (b) $\lim_{x \rightarrow 0} f(x) = 1$ $f(0)$ Undefined
 - (c) $\lim_{x \rightarrow 0} f(x) = 1$ $f(0) = 2$
 - (b) - (c) Removable discontinuity
we can remove discontinuity
by defining
 $f(0) = 1$



Discont. at (d) $x=0$
 $\lim_{x \rightarrow 0^+} f(x) = 1$ $\lim_{x \rightarrow 0^-} f(x) = 0$

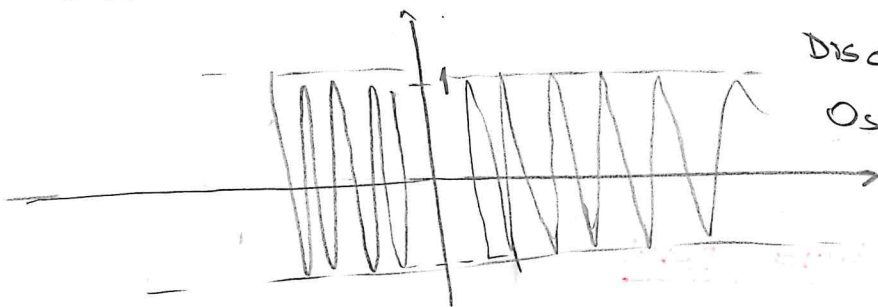
Jump discontinuity: The one-sided limits exist but they have different values



Discontinuous at $x=0$

$$\lim_{x \rightarrow 0} 1/x^2 = \infty$$

Infinite discontinuity



Discontinuous at $x=0$

Oscillating discontinuity:

it oscillates too much to have a limit as $x \rightarrow 0$

Continuity at a Point

Continuity Test

A function $f(x)$ is continuous at an interior point $x=c$ of its domain if and only if it meets the following three conditions.

- ① $f(c)$ exists (c lies in the domain of f)
- ② $\lim_{x \rightarrow c} f(x)$ exists (f has a limit as $x \rightarrow c$)
- ③ $\lim_{x \rightarrow c} f(x) = f(c)$ (the limit equals the function value)

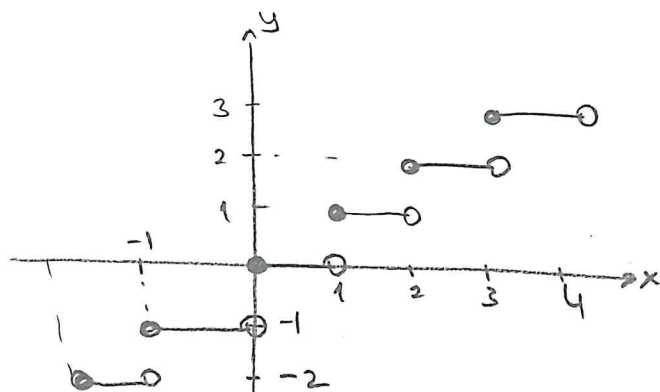
Definition:

Right and Left Continuity

We say that f is right continuous at c if $\lim_{x \rightarrow c^+} f(x) = f(c)$

We say that f is left continuous at c if $\lim_{x \rightarrow c^-} f(x) = f(c)$

Ex 7



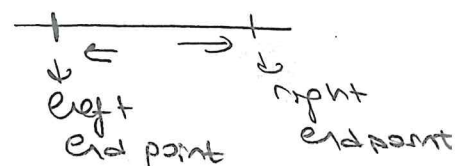
The greatest integer function is continuous at every noninteger point. It is right continuous but not left-continuous, at every integer point.

THM Function f is continuous at c if and only if it is both right continuous and left continuous at c .

Definition

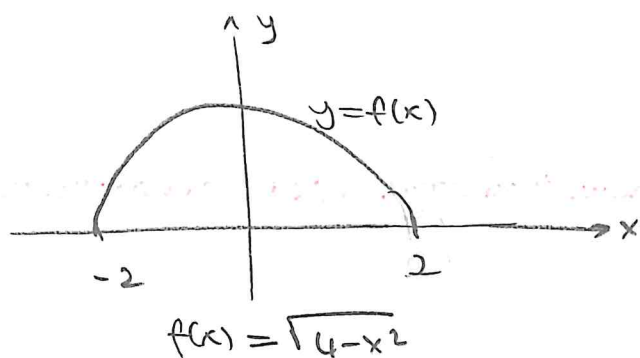
Continuity at an endpoint

→ We say that f is continuous at a left endpoint c of its domain if it is right continuous there



→ We say that f is continuous at a right point c of its domain if it is left continuous there

Ex 7



$$\lim_{x \rightarrow 2^-} f(x) = 0 = f(2)$$

$$\lim_{x \rightarrow -2^+} f(x) = 0 = f(-2)$$

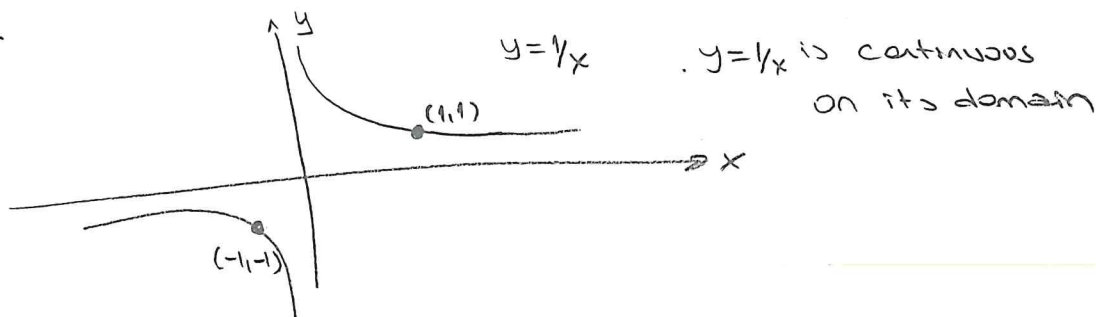
Continuous at every point of its domain.

Continuity on an Interval

Definition

Continuity on an Interval

We say that function f is continuous on the interval I if it is continuous at each point of I . In particular, we will say that f is a continuous function if f is continuous at every point of its domain.



The following functions are continuous wherever they are defined.

- (a) all polynomials
- (b) all rational functions
- (c) all rational powers $x^{m/n} = \sqrt[n]{x^m}$
- (d) the sine, cosine, tangent, secant, cosecant, cotangent functions
- (e) the absolute value function $|x|$

If f and g are both defined on an interval containing c and both are continuous at c , then the following functions are also continuous at c

- 1) $f + g$, $f - g$
- 2) $f \cdot g$
- 3) $k \cdot f$ $k \in \mathbb{R}$
- 4) f/g $g(c) \neq 0$
- 5) $(f(x))^{1/n}$ provided that $f(c) > 0$ if n is even.

THM COMPOSITE OF CONTINUOUS FUNCTIONS ARE CONTINUOUS

If $f(g(x))$ is defined on an interval containing c , and if f is continuous at L and $\lim_{x \rightarrow c} g(x) = L$, then

$$\lim_{x \rightarrow c} f(g(x)) = f(L) = f\left(\lim_{x \rightarrow c} g(x)\right).$$

In particular, if g is continuous at c ($\therefore L = g(c)$), then the composition $f \circ g$ is continuous at c .

$$\lim_{x \rightarrow c} f(g(x)) = f(g(c))$$

Continuous Extensions and Removable Discontinuities

If $f(c)$ is not defined, but $\lim_{x \rightarrow c} f(x) = L$ exists we can define a new function

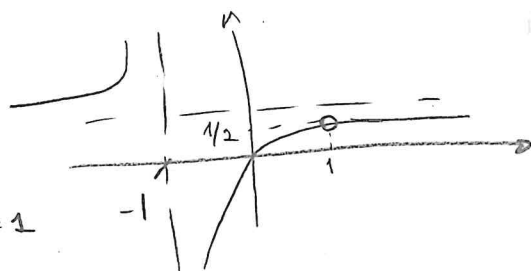
$$F(x) = \begin{cases} f(x) & \text{if } x \text{ is in the domain of } f \\ L & \text{if } x = c. \end{cases}$$

$F(x)$ is continuous at $x = c$. It is called the continuous extension of $f(x)$ to $x = c$

Ex 7 Show that $f(x) = \frac{x^2 - x}{x^2 - 1}$ has a continuous extension to $x = 1$

and find that extension

f is undefined at $x = 1$
 $f(x) = \frac{x^2 - x}{x^2 - 1} = \frac{x(x-1)}{(x+1)(x-1)} = \frac{x}{x+1}$



$F(x) = \frac{x}{x+1}$ is equal to $f(x)$ for $x \neq 1$

but is also continuous at $x = 1$

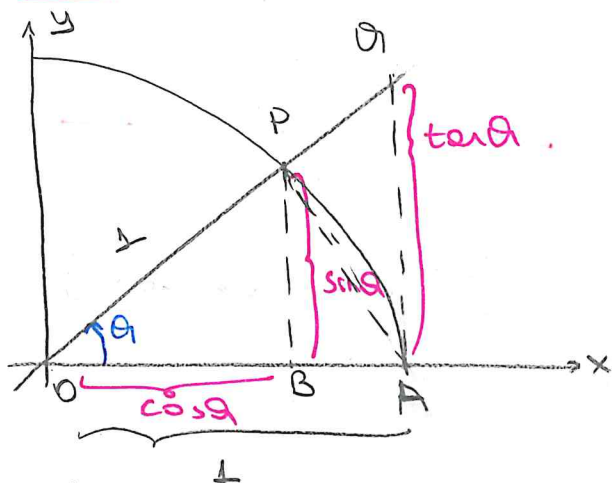
$F(x)$ has the same graph as $f(x)$, differently it involves $(1, 1/2)$

Ex 7 The function $g(x) = \begin{cases} x & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$ has removable discontinuity

at $x = 2$. To remove it define $g(2) = 2$

THM $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

PROOF



Area of $\triangle OAP \leq$ area of sector $OAP \leq$ area of $\triangle OAB$

$|OB| = \cos \theta$

$|BP| = \sin \theta$

similar triangles

$\frac{\sin \theta}{|AT|} = \frac{\cos \theta}{1} \Rightarrow |AT| = \tan \theta$

Area of $\triangle OAP = \frac{1}{2} \sin \theta$

Area of sector $OAP = \pi \cdot 1^2 \cdot \frac{\theta}{2\pi} = \frac{\theta}{2}$

Area of $\triangle OAB = \frac{1}{2} \tan \theta$

Since

Area of $\triangle AOP \leq$ area of sector $AOP \leq$ area of $\triangle AOB$

$$\frac{\sin \theta}{2} < \frac{\theta}{2} < \frac{\tan \theta}{2} \xrightarrow{\text{divide } \frac{\sin \theta}{2}} 1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta}$$

Dividing by $\frac{\sin \theta}{2}$ and reciprocal of $\cos \theta$ $\frac{\sin \theta}{\theta} < 1$ $\lim_{\theta \rightarrow 0} \cos \theta = 1$

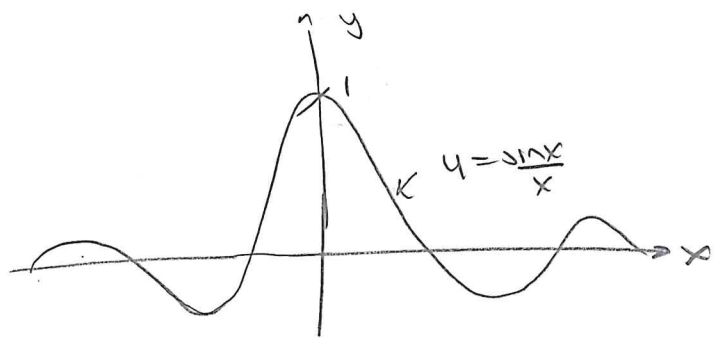
By using Squeeze theorem we obtain that

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} 1 = 1$$

Ex 7 Show that $f(x) = \frac{\sin x}{x}$ has a continuous extension to

$y=1$ and find that extension



$$F(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

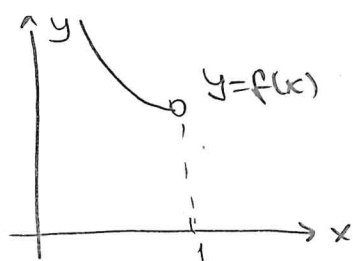
Thm The Max-Min Theorem

If $f(x)$ is continuous on the closed, finite interval $[a, b]$, then there exist numbers p and q in $[a, b]$ such that for all $x \in [a, b]$

$$f(p) \leq f(x) \leq f(q)$$

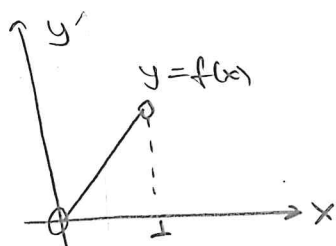
Thus, f has the absolute minimum value $m = f(p)$, taken on at the point p , and the absolute maximum value $M = f(q)$, taken on at the point q .

Continuous Functions on Closed Finite Intervals



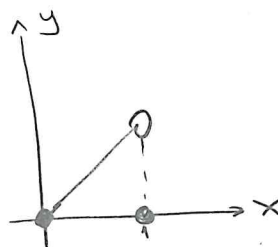
$f(x) = 1/x$ is continuous on the open interval $(0, 1)$.

It is not bounded and has neither a maximum nor a minimum value.

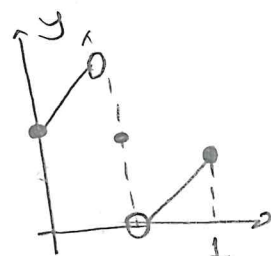


$f(x) = x$ is continuous on the open interval $(0, 1)$.

It is bounded but has neither a maximum nor a minimum value.



The function is defined on the closed interval $[0, 1]$ but is discontinuous at the endpoint $x=1$. It has a minimum value but no maximum value.

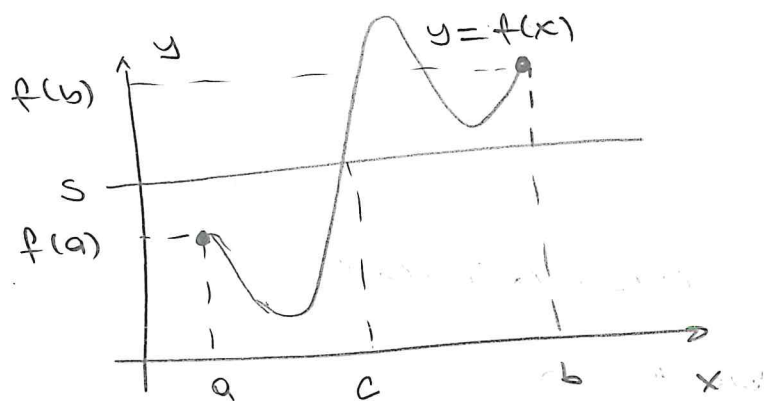


This function is discontinuous at an interior point of the closed interval. It is bounded but has neither a maximum nor a minimum.

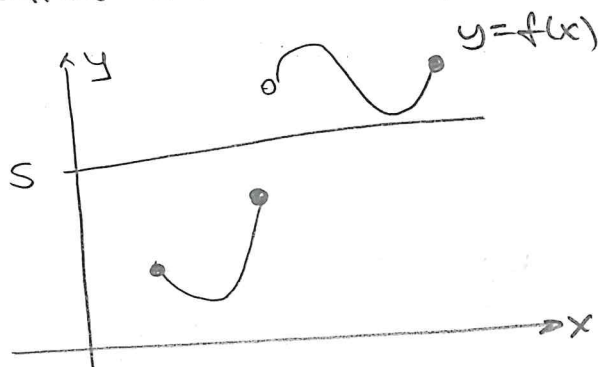
Thm: The Intermediate Theorem

If $f(x)$ is continuous on the interval $[a, b]$ and if S is a number between $f(a)$ and $f(b)$, then

there exists a number c in $[a, b]$ such that $f(c) = S$.



The continuous function f takes on the values S at some specific point c between a and b .



A discontinuous function may fail to take on all intermediate values.

Ex 7 Determine the intervals on which $f(x) = x^3 - 4x$ is positive and negative

$$f(x) = x(x^2 - 4) = x(x-2)(x+2)$$

$$f(x) = 0 \text{ at } x=0, x=2, x=-2$$

So it has a constant sign on each intervals

$(-\infty, -2), (-2, 0), (0, 2), (2, \infty)$ (If there were points a and

b in one of those intervals say $(0, 2)$ such that $f(a) < 0$

and $f(b) > 0$, then by Intermediate Value Theorem

there would exist c between a and b , and therefore

between 0 and 2 , such that $f(c) = 0$. But we know

f has no such zero in $(0, 2)$

→ To find whether $f(x)$ is positive or negative in each interval pick a value in intervals.

→ $(-\infty, -2)$, for example $x = -3$ $f(-3) = -15 < 0$ so
we have $f(x) < 0 \quad \forall x \in (-\infty, -2)$

→ $(-2, 0)$ $f(-1) = 3 > 0$ $f(x) > 0 \quad \forall x \in (-2, 0)$

→ $(0, 2)$ $f(1) = -3 < 0$ $f(x) < 0 \quad \forall x \in (0, 2)$

→ $(2, \infty)$ $f(3) = 15 > 0$ $f(x) > 0 \quad \forall x \in (2, \infty)$

Ex 7 Show that $x^3 - x - 1 = 0$ has a solution in $[1, 2]$

$$f(1) = -1 < 0$$

$$f(2) = 5 > 0$$

$$x = \frac{1+2}{2} = \frac{3}{2}$$

$$f\left(\frac{3}{2}\right) = 0.875 > 0$$

$$\left] x = \frac{3/2 + 1}{2} = \frac{5}{4} \quad f\left(\frac{5}{4}\right) < 0 \right.$$

$$x = \frac{5/4 + 3/2}{2} = \frac{11}{8}$$

Bisection method

EXAMPLES

$$\textcircled{1} \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} = \lim_{x \rightarrow 9} \frac{\cancel{\sqrt{x} - 3}}{(\cancel{\sqrt{x} - 3})(\sqrt{x} + 3)} = \frac{1}{6}$$

$$\begin{aligned} \textcircled{2} \lim_{x \rightarrow -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1} &= \lim_{x \rightarrow -1} \frac{(\sqrt{x^2 + 8} - 3)(\sqrt{x^2 + 8} + 3)}{(x + 1)(\sqrt{x^2 + 8} + 3)} \\ &= \lim_{x \rightarrow -1} \frac{(x^2 + 8) - 9}{(x + 1)(\sqrt{x^2 + 8} + 3)} = \lim_{x \rightarrow -1} \frac{\cancel{x^2 - 1}}{(x + 1)(\sqrt{x^2 + 8} + 3)} = \frac{-2}{6} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3} &= \lim_{x \rightarrow -3} \frac{(2 + \sqrt{x^2 - 5})(2 - \sqrt{x^2 - 5})}{(2 + \sqrt{x^2 - 5})(x + 3)} \\ &= \lim_{x \rightarrow -3} \frac{4 - (x^2 - 5)}{(x + 3)(2 + \sqrt{x^2 - 5})} = \lim_{x \rightarrow -3} \frac{-x^2 + 9}{(x + 3)(2 + \sqrt{x^2 - 5})} \end{aligned}$$

$$= \lim_{x \rightarrow -3} \frac{-(x^2 - 9)}{(x + 3)(2 + \sqrt{x^2 - 5})} = \lim_{x \rightarrow -3} \frac{-(x - 3)(x + 3)}{(x + 3)(2 + \sqrt{x^2 - 5})} = \frac{6}{4}$$

$$\begin{aligned} \textcircled{4} \lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}} &= \lim_{x \rightarrow 4} \frac{(4 - x)(5 + \sqrt{x^2 + 9})}{25 - x^2 - 9} = \lim_{x \rightarrow 4} \frac{(4 - x)(5 + \sqrt{x^2 + 9})}{(4 - x)(4 + x)} \\ &= \lim_{x \rightarrow 4} \frac{5 + \sqrt{x^2 + 9}}{4 + x} = \frac{10}{8} \end{aligned}$$

$$\textcircled{5} \lim_{x \rightarrow 0} \frac{|x|}{x} = ? \quad \lim_{x \rightarrow 0^+} \frac{x}{x} = 1 \quad \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1 \quad \lim_{x \rightarrow 0} \frac{|x|}{x} \text{ yphit}$$

$$\textcircled{6} \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{(x + 2)(x - 1)}{x(x - 1)} = 3$$

$$\textcircled{7} \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2} = \lim_{x \rightarrow 0} \frac{(\sqrt{x^2 + 100} + 10)(\sqrt{x^2 + 100} - 10)}{x^2(\sqrt{x^2 + 100} + 10)} = \frac{1}{2}$$

$$\textcircled{8} \lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{2 - |x|} \left(\frac{0}{0} \right) \quad \text{Around } x = -2 \text{ all numbers are negative } |x| = -x$$

$$\lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{2 + x} = \lim_{x \rightarrow -2} \frac{(x + 2)(x + 1)}{(x + 2)} = -1$$

10) for all $x \neq 0$ if $1 - \frac{x^2}{4} \leq v(x) \leq 1 + \frac{x^2}{2}$ $\lim_{x \rightarrow 0} v(x) = ?$

$\lim_{x \rightarrow 0} 1 - \frac{x^2}{4} = 1$ $\lim_{x \rightarrow 0} 1 + \frac{x^2}{2} = 1$ Sandwich Theorem $\lim_{x \rightarrow 0} v(x) = 1$

11) If x values near zero $1 - \frac{x^2}{6} \leq \frac{x \sin x}{2 - 2 \cos x} \leq 1$ $\Rightarrow$ Solved

Obtain $\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x} = ?$

$\lim_{x \rightarrow 0} 1 - \frac{x^2}{6} = 1$ $\lim_{x \rightarrow 0} 1 = 1$ Sandwich Theorem.

$\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x} = 1$

12) $f(x) = \begin{cases} ax+3 & x > 1 \\ \sqrt{x} + 2a & 0 \leq x \leq 1 \end{cases}$ If $\lim_{x \rightarrow 1} f(x)$ exists, then obtain $a = ?$

$\lim_{x \rightarrow 1^-} 1 \leq 1^+$

$\lim_{x \rightarrow 1^+} ax+3 = a+3$

$1+2a = a+3$

$\lim_{x \rightarrow 1} \sqrt{x} + 2a = 1 + 2a$
 $\sqrt{a} = 2$

13) $f(x) = \begin{cases} 3-x & x < 2 \\ \frac{x}{2} + 1 & x > 2 \end{cases}$ $\lim_{x \rightarrow 2} f(x) = ?$

$\lim_{x \rightarrow 2^-} 2 \leq 2^+$

$\lim_{x \rightarrow 2^-} f(x) = 1$

$\lim_{x \rightarrow 2^+} f(x) = 2$

$\lim_{x \rightarrow 2} f(x)$ does not exist

14) $\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = ?$ $\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = \lim_{\theta \rightarrow 0} \frac{1}{\frac{\sin \theta}{\theta}} = 1$

15) $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = ?$ $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\cos \theta \cdot \theta} = 1$

16) $\lim_{y \rightarrow 0} \frac{\sin 3y}{4y} = \lim_{y \rightarrow 0} \frac{3}{4} \frac{\sin 3y}{3y} = \lim_{y \rightarrow 0} \frac{3}{4} \frac{\sin 3y}{3y} = \frac{3}{4} \cdot 1 = \frac{3}{4}$

17) $\lim_{x \rightarrow 0} \frac{x^2 - x + \sin x}{2x} = \lim_{x \rightarrow 0} \left(\frac{x}{2} - \frac{1}{2} + \frac{\sin x}{2x} \right)$
 $= \lim_{x \rightarrow 0} \left(\frac{x^0}{2} - \frac{1}{2} + \frac{1}{2} \frac{\sin x}{x} \right) = -\frac{1}{2} + \frac{1}{2} = 0$

$$(18) \lim_{h \rightarrow 0} \frac{\sin(\sinh)}{\sinh} = \lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$$

$$h \rightarrow 0 \quad \sinh \rightarrow 0$$

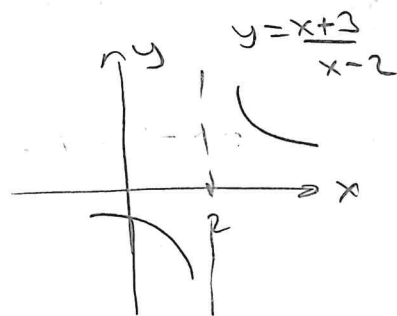
$$(19) \lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 3}{3x^2 + 5} = \lim_{x \rightarrow \infty} \frac{x^2(2 - \frac{3}{x} + \frac{3}{x^2})}{x^2(3 + \frac{5}{x^2})} = \frac{2}{3}$$

$$(20) \lim_{x \rightarrow \infty} \frac{5x+2}{2x^3-1} = \lim_{x \rightarrow \infty} \frac{x(5 + \frac{2}{x})}{x^3(2 - \frac{1}{x^3})} = 0$$

$$(21) \lim_{x \rightarrow \infty} \sin x = ? \quad \text{Limit doesn't exist because } -1 \leq \sin x \leq 1 \text{ } \sin x \text{ oscillates in } [-1, 1] \text{ interval.}$$

$$(22) \lim_{x \rightarrow 2} \frac{x+3}{x-2} = ? \quad \lim_{x \rightarrow 2^+} \frac{x+3}{x-2} = \infty$$

$$\lim_{x \rightarrow 2^-} \frac{x+3}{x-2} = -\infty$$



Limit does not exist.

$$(23) \lim_{x \rightarrow \infty} \frac{-1}{x^2(x+1)} = -\infty$$

$$(24) \lim_{x \rightarrow \infty} \sqrt{\frac{8x^2-3}{2x^2+x}} = \lim_{x \rightarrow \infty} \sqrt{\frac{x^2(8-\frac{3}{x^2})}{x^2(2+\frac{1}{x})}} = \sqrt{4} = 2$$

$$(25) \lim_{x \rightarrow \infty} \frac{2+\sqrt{x}}{2-\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}(\frac{2}{\sqrt{x}} + 1)}{\sqrt{x}(\frac{2}{\sqrt{x}} - 1)} = \frac{1}{-1} = -1$$

$$(26) \lim_{x \rightarrow 2^+} \frac{8}{x^2-4} = \frac{2}{x-2} \quad (\infty - \infty)$$

$$\lim_{x \rightarrow 2^+} \frac{8-2(x+2)}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{8-2x-4}{(x-2)(x+2)} = \lim_{x \rightarrow 2^+} \frac{-2(x-2)}{(x-2)(x+2)} = \frac{-1}{2}$$

$$(27) \lim_{x \rightarrow \infty} (\sqrt{1+x^2} - \sqrt{2+5x+x^2}) \quad (\infty - \infty)$$

Multiply and divide with $\sqrt{1+x^2} + \sqrt{2+5x+x^2}$ [rationalize]

$$\lim_{x \rightarrow \infty} \frac{1+x^2-2-5x-x^2}{\sqrt{1+x^2} + \sqrt{2+5x+x^2}} = \lim_{x \rightarrow \infty} \frac{-1-x-5}{\sqrt{\frac{1}{x^2}+1} + \sqrt{\frac{2}{x^2}+\frac{5}{x}+1}} = \frac{-5/2}{1+1} = -\frac{5}{4}$$

(28) $\lim_{x \rightarrow \pi/2} \frac{\sin(x - \pi/2)}{x - \pi/2} = ?$ ($\frac{0}{0}$) indetermined form

$x \rightarrow \pi/2$ $x - \pi/2 \rightarrow 0$ $\lim_{x \rightarrow \pi/2} \frac{\sin(x - \pi/2)}{x - \pi/2} = \lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$

(29) $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \pi/2} = ?$ ($\frac{0}{0}$) $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \pi/2} = \lim_{x \rightarrow \pi/2} \frac{-\sin(x - \pi/2)}{x - \pi/2} = -1$

$\cos x = \sin(\pi/2 - x) = -\sin(x - \pi/2)$

complementary identity sine function is odd.

(30) $\lim_{x \rightarrow 0} x^3 \cos 1/x = ?$

$-1 \leq \cos 1/x \leq 1 \Rightarrow -x^3 \leq x^3 \cos 1/x \leq x^3$ $\lim_{x \rightarrow 0} \mp x^3 = 0$

Sandwich theorem $\lim_{x \rightarrow 0} x^3 \cos 1/x = 0$

(31) $\lim_{x \rightarrow -\infty} \frac{x^2 (\sin x + \cos^3 x)}{(x^2 + 1)(x - 3)} = \lim_{x \rightarrow -\infty} \frac{x^2 (\frac{\sin x}{x} + \frac{\cos^3 x}{x})}{x^2 (1 + \frac{1}{x^2}) (1 - \frac{3}{x})} = 0$

Sandwich Thm $-1 \leq \sin x \leq 1 \Rightarrow -1/x \leq \frac{\sin x}{x} \leq 1/x$

$-1 \leq \cos^3 x \leq 1 \Rightarrow -1/x \leq \frac{\cos^3 x}{x} \leq 1/x$

$\lim_{x \rightarrow -\infty} -1/x = \lim_{x \rightarrow -\infty} 1/x = 0$

$\lim_{x \rightarrow -\infty} \frac{\sin x}{x} = \lim_{x \rightarrow -\infty} \frac{\cos^3 x}{x} = 0$

$\lim_{x \rightarrow -\infty} \frac{\frac{\sin x}{x} + \frac{\cos^3 x}{x}}{(1 + 1/x^2)(1 - 3/x)} = 0$

(32) $\lim_{x \rightarrow \pi/2^-} (x - \pi/2) \tan x = ?$ ($0 \cdot \infty$ indetermined form)

$\lim_{x \rightarrow \pi/2^-} (x - \pi/2) \frac{\sin x}{\cos x} = \lim_{x \rightarrow \pi/2^-} \frac{x - \pi/2}{\cos x} \cdot \sin x$

$\lim_{x \rightarrow \pi/2^-} \sin x = 1$ $\lim_{x \rightarrow \pi/2^-} \frac{x - \pi/2}{\cos x} = \lim_{u \rightarrow 0} \frac{u}{-\sin(\pi/2 - u)} = -1$ $\lim_{x \rightarrow \pi/2^-} (x - \pi/2) \tan x = -1$

33 For the following functions obtain a and b values if they are continuous.

(a) $f(x) = \begin{cases} a^2x - 2a & x \geq 2 \quad a > 0 \\ 12 & x < 2 \end{cases}$

(b) $g(x) = \begin{cases} \frac{x-b}{b+1} & x < 0 \\ x^2 + b & x \geq 0 \end{cases}$

(a) $a^2x - 2a = p(x)$ is a polynomial and continuous at every point.

$q(x) = 12$ similar to $p(x)$ it is also continuous.

Only point that must be considered is 2.

$\lim_{x \rightarrow 2^+} a^2x - 2a = 2a^2 - 2a \quad f(2) = 2a^2 - 2a$
 $\lim_{x \rightarrow 2^-} 12 = 12$

To have limit $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$

$2a^2 - 2a = 12 \quad a^2 - a - 6 = 0 \quad (a-3)(a+2) = 0$
 $a = 3 \quad a = -2$

$a > 0$ we want so $a = 3$.

(b) Only critical point is 0.

$\lim_{x \rightarrow 0^+} f(x) = 0 + b$

$\lim_{x \rightarrow 0^-} \frac{x-b}{b+1} = \frac{-b}{b+1}$

$b = \frac{-b}{b+1}$

$b(b+1) = -b$

$b^2 + b = -b$

$b^2 + 2b = 0$

$b(b+2) = 0$

$b = 0$ or $b = -2$ will

be enough.

